

English
Medium



EDU TERIA

72nd BPSC

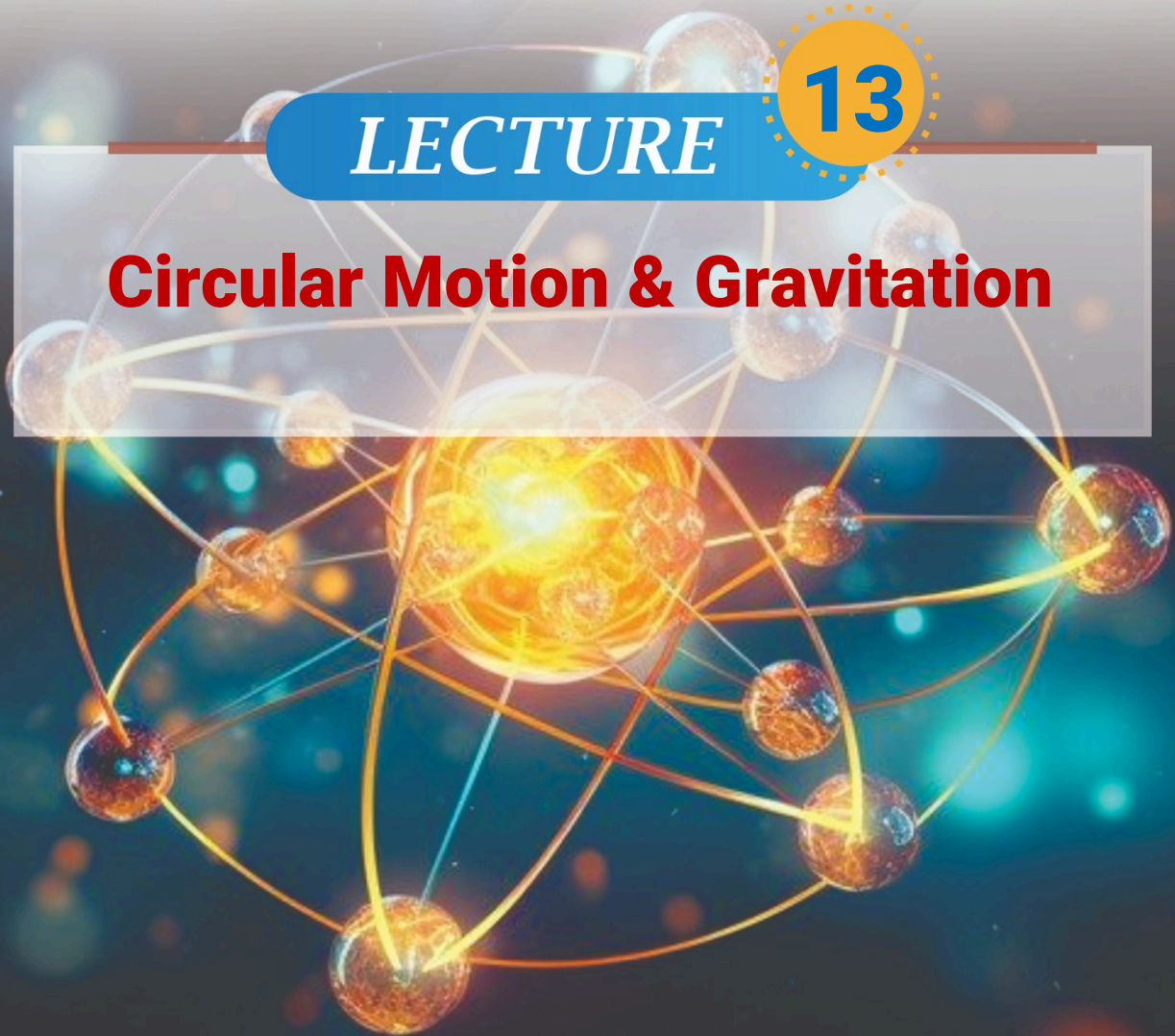
Prelims Online Batch

PHYSICS

LECTURE

13

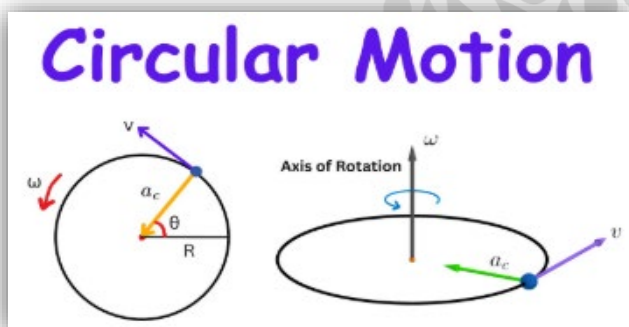
Circular Motion & Gravitation



Circular Motion & Gravitation

Circular motion

- If a particle moves in a plane such that its distance from a fixed (or moving) point remains constant, then its motion is called as circular motion with respect to that fixed or moving point.
- That fixed point is called the Centre, and the corresponding distance is called the radius of circular path.



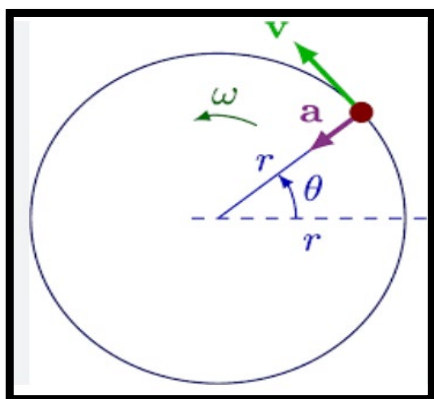
Types of Circular Motion

I. Uniform Circular Motion

- If a particle moves with constant speed. It means net external force applied is zero. This type of motion is termed as uniform circular motion.
- In uniform circular motion tangential acceleration is zero. Non-uniform circular motion = $\frac{(V_2 - V_1)}{t} = 0$
- Motion of variable speed due to tangential acceleration known as non-uniform circular motion.

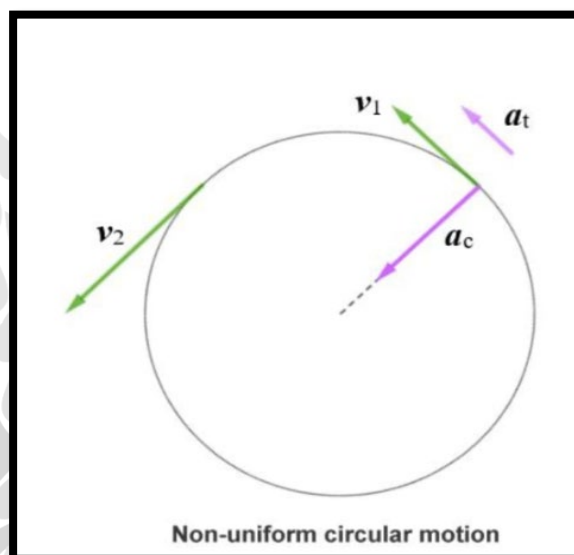
E. g- Circular motion in vertical plane.

- Essential force required for circular motion known as centripetal force.



II. Non-Uniform Circular Motion

Motion of a body along a circular path in which its speed (magnitude of velocity) changes with time, and hence its velocity changes due to both change in magnitude and direction.



- Example:** a car accelerating or slowing down on a circular track

Types of Acceleration in Non-Uniform Circular Motion

1. Centripetal (Radial) Acceleration :

- Acts toward the center of the circle Responsible for change in direction
- Always perpendicular to velocity

$$a_c = \frac{v^2}{r}$$

2. Tangential Acceleration :

- Acts along the tangent of the circle
- Responsible for change in speed Can be positive (speed increasing) or negative (speed decreasing)

$$a_t = \frac{dv}{dt}$$

Total Acceleration

Since both components are perpendicular, total acceleration is:

$$a = \sqrt{a_c^2 + a_t^2}$$

**Terms related to Circular Motion**

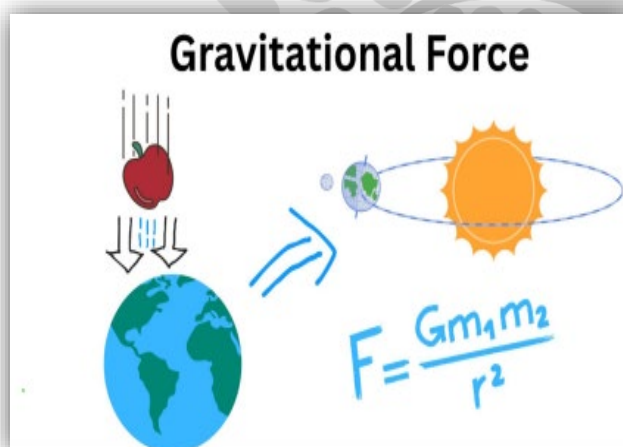
- **Angular Displacement:** Angle traced by the position vector of a particle moving w. r. t. some fixed point is called angular displacement.
- **Frequency (n):** Number of revolutions described by particle per second is its frequency. Its unit is revolutions per second (rps) or revolutions per minute (rpm).
 $n = 1/T$, $T = \text{Time period}$
- **Time Period (T):** It is the time taken by particle to complete one revolution
- **Angular Velocity (w):** It is defined as the rate of change of angular displacement of a moving particle, w. r. t. to time.

Gravitation

Gravitation is the universal force of attraction that exists between any two bodies in the universe having mass, acting along the line joining their centers.

The discovery of the law of gravitation

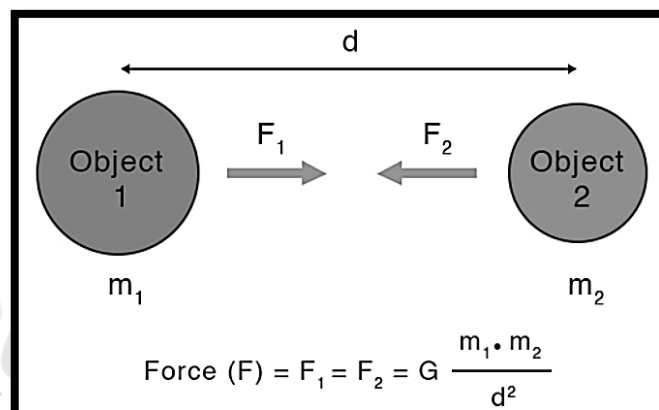
- The way the law of universal gravitation was discovered is often considered as the paradigm of modern scientific technique.

**The discovery of the law of gravitation**

- ❑ The major steps involved were:
- The hypothesis about planetary motion given by Nicolaus Copernicus (1473-1543).
 - The careful experimental measurements of the positions of the planets and the Sun by Tycho Brahe (1546-1601).
 - Analysis of the data and the formulation of empirical laws by Johannes Kepler (1571-1630).
 - The development of a general theory by Isaac Newton (1642-1727).

Newton's law of Gravitation

- It states that every particle in the universe attracts all other particles with a force which is directly proportional to the product of their masses and is inversely proportional to the square of the distance between them.

**Universal Gravitational Constant "G"**

- Universal Gravitational constant is a scalar quantity. Its value is same throughout the universe.
- G does not depend on the nature and size of universe. It does not depend even upon the nature of the medium between the bodies.
- Its value was first found by the scientist "Henry Cavendish" with the help of "Torsion Balance" experiment.

Gravitational force is always attractive

- Gravitational forces are developed in the form of action and reaction pairs.
- Hence, they obey Newton's third law of motion.
- It is independent of the nature of medium between two masses.
- Gravitational forces are central forces as they act along the line joining the Centre of gravity of the two bodies
- Gravitational forces are conservative forces, so work done by gravitational force does not depend on path.
- If any particle moves along a closed path under the action of BPSC PRELIMS gravitational force, then the work done by this force is always zero round the trip.
- Gravitational force is weaker than the electromagnetic and nuclear forces.
- Force developed between any two masses is called gravitational force and force between Earth and any body is called force of gravity.



**ACCELERATION DUE TO GRAVITY**

- In Newton's law of gravitation, the force of attraction between any two bodies is gravitation.
- If one of the bodies is Earth, then gravitation is called 'gravity'.
- Hence, gravity is the force by which Earth attracts a body towards its Centre.
- It is a special case of gravitation.

Value of 'g' does not depend upon:

- (i) shape and size of object
- (ii) Mass of object

Dependency of 'g':

- (i) Mass of earth/planet
- (ii) Radius of earth/ planet

ACCELERATION DUE TO GRAVITY

- Let us assume that Earth is a uniform sphere of mass M and radius R.
- The magnitude of the gravitational force of Earth on a particle of mass m, located outside the Earth at a distance r from its Centre, is

$$F = \frac{GMm}{r^2}$$

where:

- G= universal gravitational constant
- M= mass of the Earth
- m= mass of the particle
- r= distance from the center of the Earth

Now according to Newton's second law

Any value 'g' measured at a given location will differ from the 'g' value calculated according to equation due to three reasons:

1. Earth's mass is not distributed uniformly.
2. Earth is not a perfect sphere and
3. Earth rotates.

